

The Metropolitan Water Reclamation District of Greater Chicago (MWRD) reduces urban stormwater runoff through interventions such as downspout disconnection on large institutional rooftops. Identifying strong candidates requires sifting through large, fragmented parcel and rooftop datasets. To help MWRD planners prioritize sites, the team developed an interactive screening tool that narrows thousands of buildings to focused candidates.

The team identified tax-exempt parcels, such as schools, churches, and government buildings because these properties tend to have large, flat rooftops suited for stormwater interventions. These parcels were linked to rooftop geometries and filtered for eligibility. The team also estimated whether rooftops had nearby open green space, since downspout disconnection requires permeable ground nearby to absorb redirected rainwater.

We created a scalable, map-based tool that transforms complex spatial data into an accessible planning resource. In Oak Park, the system reduced hundreds of rooftops to a smaller subset of high-interest sites. We learned that most had small rooftops, but a few large ones represented disproportionate impact opportunities (Figure 1). Using this tool, MWRD can prioritize which buildings to review first.

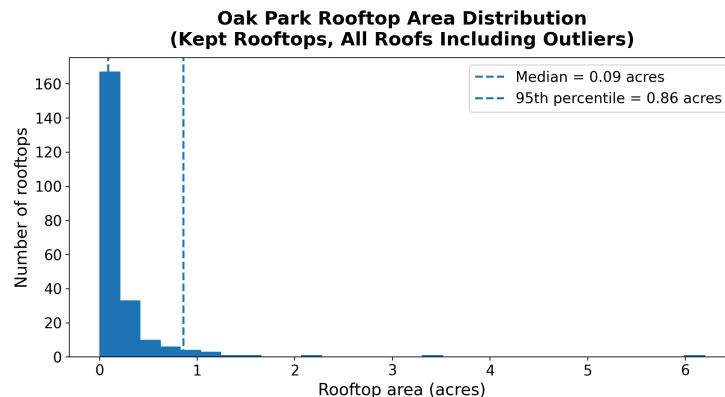


Figure 1: Oak Park rooftop size distribution with median & 95th percentile markers

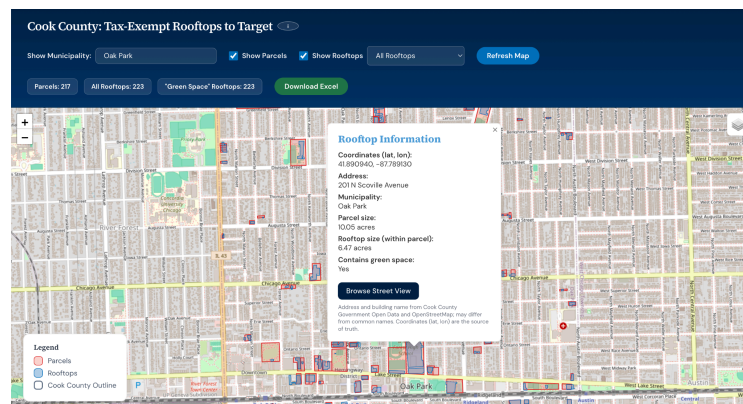


Figure 2: Interactive web tool for filtering and exploring candidate rooftops by municipality